

Indus Valley Script Research Project

Corpus, Positional Grammar, and Hypothesis Control

Research Lead: Codex
Supervisor: Divyanshu

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Abstract

This document records a disciplined research program for the Indus Valley sign system. The project begins from corpus hygiene, sign distribution, directionality, and rival-model testing rather than from proposed readings. Linguistic hypotheses, including Dravidian, Indo-Aryan/Indo-Iranian, Munda-like, isolate, and mixed administrative models, are admitted only when they produce testable predictions.

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1 Project Scope

The aim of this project is not to announce a decipherment prematurely. The aim is to build the strongest possible evidence pipeline: corpus hygiene, sign statistics, positional grammar, non-linguistic baselines, and only then constrained linguistic hypotheses.

1.1 Working Stance

- Treat the Indus system as undeciphered.
- Treat all proposed readings as hypotheses, not data.
- Prefer reproducible tests over attractive resemblances.
- Compare Dravidian, Indo-Aryan/Indo-Iranian, Munda-like, isolate, and non-linguistic models under the same evidentiary rules.
- Separate graphemic questions from linguistic questions: first ask what the signs are doing, then ask whether speech is involved.

1.2 Current Data

- `data/ivs_corpus_cleaned.csv`: initial neutral working corpus.
- `data/ivs_corpus.csv`: same row count, but includes proposed Sanskrit readings and translations. These fields are excluded from the baseline because they risk circular reasoning.

1.3 Research Phases

1. Corpus audit and provenance.
2. Sign inventory normalization.
3. Directionality and positional class analysis.
4. Formula, n-gram, and network analysis.
5. Non-linguistic baseline modeling.
6. Linguistic hypothesis testing.
7. Narrow, falsifiable candidate readings only where distributional evidence permits them.

1.4 Evidence Ladder

Speculative Plausible idea, not yet tested.

Suggestive Agrees with some observed pattern but has weak alternatives.

Testable Predicts a measurable corpus pattern.

Statistically supported Survives predefined quantitative tests.

Philologically plausible Agrees with linguistic, archaeological, and comparative constraints.

No claim enters the project as a conclusion. It enters as a hypothesis and earns its place.

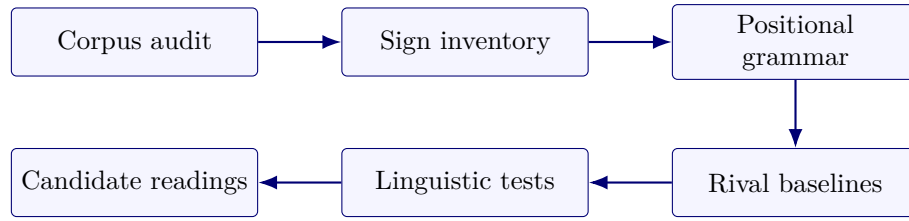


Figure 1: Research pipeline. Candidate readings are deliberately last.

2 Methodology Charter

2.1 North Star

The project will pursue decipherment only through constraints. We will not begin by assigning phonetic or semantic values to signs. We will begin by asking what the corpus permits.

2.2 Core Rules

1. The cleaned corpus is the initial data layer. Claimed readings and translations are hypothesis material, not evidence.
2. Any language-family claim must predict distributional behavior before it is used to explain individual signs.
3. Iconographic resemblance is not a reading. It can generate a hypothesis, but not validate one.
4. A proposed value must be tested against more inscriptions than the one that inspired it.
5. Negative evidence matters: failed hypotheses stay in the ledger so we do not rediscover them under new names.
6. We distinguish at least four levels: sign, allograph, sign cluster, and artifact formula.
7. Archaeological context is part of the grammar. Site, medium, object type, iconography, completeness, and directionality are not metadata afterthoughts.

2.3 Data Tiers

Tier A Complete inscriptions with known or strongly supported directionality, clean sign sequence, stable artifact metadata.

Tier B Complete or near-complete inscriptions with uncertain directionality or minor damage.

Tier C Damaged, fragmentary, unusual, or contextually uncertain inscriptions.

Tier D Disputed, unprovenanced, translated-only, or otherwise circular records. These are excluded from baseline analysis.

2.4 Initial Research Questions

- Which signs are strongly initial, medial, or terminal?
- Which signs behave like numerals, classifiers, titles, commodities, names, or formula delimiters?
- Do object types have distinct sign grammars?

- Does the system look more like language, restricted writing, emblematic identifiers, administrative coding, or a mixed system?
- Do any observed structures resemble Dravidian or Indo-Aryan morphology strongly enough to produce testable predictions?

2.5 Decision Rule for Candidate Readings

A reading is not allowed into the main model unless it satisfies all three conditions:

1. It explains a distributional fact in the corpus.
2. It predicts at least one additional distributional fact not used to invent it.
3. It is more economical than credible non-linguistic and alternative linguistic explanations.

2.6 Bias Controls

- Maintain rival models in parallel.
- Score claims by evidence level.
- Prefer blind tests: train on one site or object class, test on another.
- Keep sign-number mappings distinct across Mahadevan, Parpola, Wells, and local datasets.
- Document every normalization step.

2.7 Computational-First Protocol

Because the project is led from a computer-science and data-science standpoint, claims should be expressed as scripts, features, scores, simulations, and reproducible outputs wherever possible. Linguistic ideas are useful when they become model constraints. The preferred workflow is:

1. Convert an intuition into measurable variables.
2. Generate corpus-wide features rather than isolated examples.
3. Use simulations, resampling, or held-out tests when metadata permits.
4. Treat model output as a ranking of hypotheses, not as proof.
5. Promote a hypothesis only when the code, the corpus, and the visual/artifact record agree.

Comparisons with Brahmic scripts are therefore allowed as hypothesis generators, especially for base-sign and modifier behavior, but not as assumed ancestry. A proposed Brahmic-style mechanism must predict a measurable pattern in the Indus corpus before it can influence interpretation.

3 Source Register

This register tracks sources we may use and how much evidentiary weight they should carry.

Source	Use	Status	Notes
Local data/ivs_corpus_cleaned.csv	Initial working corpus	Active, provenance to verify	5,679 rows; excludes claimed Sanskrit readings/translations.
Local data/ivs_corpus.csv	Hypothesis archive only	Restricted	Includes proposed Sanskrit readings/translations; not baseline data.
Yajnadevam lipi repository [Yajnadevam, 2026]	Public source match for local CSV structure	Active provenance lead	Exposes an inscriptions.csv stream whose fields and sample rows align with the local corpus.
Fuls/ICIT documentation [Fuls, 2026]	Directionality and text-code conventions	Active provenance lead	Documents the interactive corpus and its R/L text-code search conventions.
Fuls, “Positional Analysis of Indus Signs” [Fuls, 2013]	Positional method and comparison panels	Tracked local PDF	Stored at references/pdfs/fuls/Fuls_2013_Posit PDF page 10 contains a panel for 692, closing the missing visual side for the 692/920 Tier 1 pair.
Fuls, <i>Corpus of Indus Inscriptions</i> [Fuls, 2022]	ICIT corpus and sign-list reference	Tracked local PDF	Stored at references/pdfs/fuls/Fuls_2022_Corpu PDF page 21 contains a high-number sign-list panel covering ICIT 703–958, including 705, 706, 817, 820, 861, and 920.
Harappa catalog review [Harappa.com, 2026]	Public description of Fuls’ sign catalog	Contextual reference	Confirms that the catalog is visually detailed and designed as a sign-reference tool, but does not provide a machine-readable target-code image table.
Deutsche Digitale Bibliothek record [Deutsche Digitale Bibliothek, 2026]	Bibliographic access to Fuls’ catalog	Contextual reference	Confirms publication details for the 560-page illustrated catalog, while noting that only the table of contents is digitally open.
Local Fuls 2023 SearchInside PDF sample	Supplementary ICIT visual mapping	Tracked local PDF	Stored at references/pdfs/fuls/Fuls_2023_Catal The sample independently shows additional catalog visuals, including 820 and 920.
Daggumati and Revesz [Daggumati and Revesz, 2021]	Allograph-review method	Active methodological reference	Provides a cautious framework for testing redundancies, mirroring, directionality, space-saving, catalog errors, and image-based allograph claims.

Source	Use	Status	Notes
Mahadevan, <i>The Indus Script: Texts, Concordance and Tables</i> [Mahadevan, 1977]	Standard concordance reference	Foundational; tracked local PDF	Stored at references/pdfs/mahadevan/Mahadevan_1 Often cited as M77; sign counts and text lines differ depending on filtering.
Joshi and Parpola, <i>Corpus of Indus Seals and Inscriptions</i> , Vol. 1 [Joshi and Parpola, 1987]	Artifact-image authority	Foundational; tracked local PDF	Stored at references/pdfs/cisi/Joshi_Parpola_1 Useful later for seal/inscription image checks once ICIT/CISI/page links are built.
Marshall, <i>Mohenjo-daro and the Indus Civilization</i> , Vol. II [Marshall, 1931]	Excavation context and early text discussion	Tracked local PDF	Stored at references/pdfs/marshall/Marshall_19 Potentially useful for archaeological context and older readings, not a baseline sign-list authority.
CISI remaining volumes	Corpus/reference tradition	Candidate future sources	Additional volumes may be needed for Pakistan and later-discovered materials.
mayig indus-valley-script-computer	Open digital CISI-computed corpus	Candidate comparative corpus	WIP digitization of CISI in JSON; records graphemes left-to-right while assuming right-to-left reading for the script.
Yadav et al. [Yadav et al., 2010]	N-gram/statistical baseline	Active methodological reference	Reports Zipf-Mandelbrot behavior, positional asymmetry, strong bigram correlations, and Markov modeling.
Rao et al. [Rao et al., 2009b]	Conditional entropy comparison	Active methodological reference	Argues Indus conditional entropy resembles linguistic systems more than tested non-linguistic controls.
Rao et al. [Rao et al., 2009a]	Markov modeling	Active methodological reference	Uses positional statistics and Markov methods for syntax and missing-sign prediction.
Farmer, Sproat, and Witzel [Farmer et al., 2004]	Non-linguistic critique	Required adversarial model	Argues against a linguistic-script thesis and stresses brevity, lack of long texts, and non-linguistic symbol-system parallels.
Sinha et al. [Sinha et al., 2010]	Network analysis	Candidate methodological reference	Reports network patterns suggestive of syntactic organization.
Nair [Nair, 2026]	Recent synthetic non-linguistic baseline	Provisional	New preprint; useful as an emerging baseline design, not yet treated as settled.

3.1 Immediate Source Tasks

1. Identify the provenance of the two local CSV files.
2. Determine the sign numbering system used locally.
3. Build concordance crosswalks where possible: local code, Mahadevan, Parpola, Wells.
4. Build a local PDF page index for the curated reference library.
5. Link ICIT/CISI identifiers to artifact-image pages in the local CISI volume.
6. Decide whether to import the open JSON corpus as a second corpus layer.

4 Initial Corpus Profile

Generated from `data/ivs_corpus_cleaned.csv` on 2026-05-15.

4.1 Basic Counts

Measure	Value
Rows	5,679
Rows marked complete	3,673
Unique sites	77
Unique artifact types	32
Analyzable sign tokens	18,044
Unique analyzable sign codes	712
Eroded 000 tokens	1,881
Encoded-space 999 tokens	17
Mean parsed text length	3.512 signs

Table 2: Initial corpus counts from the first profiling pass.

4.2 Length Distribution

4.3 Directionality Metadata

4.4 Top Analyzable Sign Codes

4.5 Immediate Interpretation

The local corpus is large enough for statistical analysis, but the 712-code analyzable inventory is too broad to treat as a simple primary-sign inventory. The next step is to separate allographs, numerals, uncertain signs, damaged signs, encoded spaces, and local sign-list conventions before making claims about writing-system type or language family.

5 Hypothesis Ledger

This ledger records live hypotheses. It is deliberately conservative: hypotheses may be beautiful, but they must become testable.

Parsed length	Rows
1	698
2	1,514
3	1,051
4	863
5	669
6	381
7	285
8	102
9	55
10	26
11	22
12	4
13	7
16	1
17	1

Table 3: Parsed inscription length distribution in the cleaned corpus.

Direction	Rows
R/L	4,265
–	776
NR	386
L/R	215
T/B	16
BUS	10
SYM	8
R/L with trailing space	3

Table 4: Directionality values in the cleaned corpus.

Sign	Count	Sign	Count
740	1,928	861	272
002	851	003	259
032	588	031	258
700	578	235	250
033	523	820	244
400	502	590	228
220	477	741	225
240	354	060	222
520	321	817	221
390	290	705	219

Table 5: Top 20 analyzable sign codes in the profiling pass.

ID	Hypothesis	Current Level	First Test
H0	The system is primarily non-linguistic: emblematic, administrative, cultic, or owner/lineage marking.	Testable	Compare brevity, repetition, positional rigidity, and object-class specificity against non-linguistic baselines.
H1	The system is restricted writing: linguistic or para-linguistic formulae, not full prose.	Testable	Look for stable positional classes, formula slots, and controlled alternations across sites/media.
H2	The system is logo-syllabic or logo-phonetic with many allographs.	Speculative	Test whether sign inventory can be compressed into a smaller functional inventory without losing positional structure.
H3	A Dravidian substrate explains some sign clusters or terminal elements.	Speculative	Search for suffix-like positional behavior and compare only after corpus classes are established.
H4	Indo-Aryan or Indo-Iranian readings explain some sign clusters.	Speculative	Require chronology-sensitive linguistic forms and predictions beyond isolated lexical resemblance.
H5	The corpus reflects a multilingual administrative ecology using language-neutral signs plus limited phonetic elements.	Speculative	Test whether regional/site differences cluster by artifact function more than by putative language.
H6	Numeral-like signs and commodities are the easiest first partial decipherment target.	Suggestive	Isolate stroke/numeral candidates, following-sign distributions, and correlations with object types, weights, or measures.

5.1 First Decision

The first serious analysis should not be a reading. It should be a positional grammar:

1. Normalize directionality.
2. Extract start, end, and medial sign distributions.
3. Split by artifact type and site.
4. Identify signs whose positional entropy is unusually low.
5. Use those signs to define candidate functional classes.

6 Provenance and Directionality Note

6.1 Corpus Provenance

The local CSV fields and sample records match the public `lipi` corpus layout and data stream exposed in the Yajnaddevam repository [Yajnaddevam, 2026]. That resource appears to derive from or align with the Interactive Corpus of Indus Texts (ICIT) tradition associated with Wells and Fuls [Fuls, 2026]. This is a working provenance statement, not yet a final bibliographic claim.

6.2 Text-Code Order

The local `text` field uses sign codes in a plus-delimited format such as `+740-540-002-820+`. ICIT documentation labels the searchable text field as text in R/L order. For internal analysis, this project distinguishes:

Text-code order The sequence exactly as represented in the CSV/ICIT code field.

Normalized internal reading order The sequence reversed from text-code order so that analysis runs from initial side to terminal side.

The positional script now preserves both modes. The default mode is normalized internal reading order; `-UseTextCodeOrder` emits the raw text-code control report.

6.3 Practical Consequence

The directionality control shows that initial and terminal classes invert between the two modes, while medial signs remain largely stable. Therefore, claims about prefixes or endings must explicitly state which direction convention is being used. Medial behavior is the safer first target for sign-class analysis.

7 First Findings: Positional Grammar Pass

Generated on 2026-05-15 from the positional-analysis outputs.

7.1 Scope

This pass used a stricter subset of the cleaned corpus:

- complete inscriptions only;
- directions limited to R/L and L/R;
- rows containing 000 excluded;
- the ICIT-style text-code order was compared against a normalized internal reading order.

Rows analyzed: 3,152. Unique analyzable sign codes analyzed positionally: 605.

7.2 Early Functional Classes

The following are behavioral classes, not readings.

7.2.1 Strong End Candidates

Sign	Total	Terminal percent
400	360	90.28
527	47	87.23
151	71	85.92
520	255	83.53
700	448	80.13
090	125	77.60
156	92	77.17
154	38	76.32
740	1,361	67.89

Table 7: Strong terminal candidates in normalized internal reading order.

This suggests a terminal class that may include formula closers, titles, classifiers, commodities, or grammatical endings.

7.2.2 Strong Start Candidates

Sign	Total	Initial percent
501	31	93.55
817	164	89.02
034	132	84.85
692	54	79.63
820	165	75.76
503	69	75.36
413	39	74.36
861	212	61.32
003	195	59.49

Table 8: Strong initial candidates in normalized internal reading order.

This suggests an opening class that may include prefixes, header signs, owner/title markers, or object-class indicators.

7.2.3 Strong Medial Candidates

Sign	Total	Medial percent
845	48	100.00
752	49	97.96
060	181	97.79
760	90	97.78
585	47	95.74
500	20	95.00
002	593	94.94
100	98	94.90
741	170	88.82

Table 9: Strong medial candidates in normalized internal reading order.

This suggests a medial class that may carry core lexical, classifying, or internal-formula functions.

7.3 Directionality Finding

The first directionality diagnostic produced an important control result: start and end classes invert when analyzed in raw text-code order, while medial signs remain stable. The ICIT source convention appears to encode text in R/L code order, where the terminal side is on the left of the code and the initial side is on the right. Therefore, internal linguistic analysis should reverse text-code order and work from initial to terminal.

7.4 Next Test

The next analytical milestone is sign-inventory compression. We must separate primary signs, allographs, stroke/numeral candidates, damaged signs, encoded spaces, and local sign-list conventions before treating the 712 observed analyzable codes as linguistic units.

8 Sign-Inventory Compression

8.1 Rationale

The inventory pass treats every observed code as a code-level observation, not as a final graphemic unit. This distinction is necessary because the local corpus contains 712 analyzable sign codes after excluding eroded 000 and encoded-space 999. ICIT documentation describes the Wells list as a 676-sign sign list [Fuls, 2026], while also documenting function and spacing conventions. The discrepancy does not by itself prove that the local corpus has extra signs; it tells us that a crosswalk and variant audit must precede any claim about the functional size of the script.

8.2 Inventory Overview

The script `scripts/sign-inventory-analysis.ps1` generated a code-level inventory from `data/ivs_corpus_cleaned.csv`. The output files are:

- `outputs/sign_inventory_stats.csv`;
- `outputs/sign_frequency_bands.csv`;
- `outputs/sign_coverage_thresholds.csv`;

- `outputs/sign_candidate_classes.csv`;
- `outputs/sign_inventory_analysis.tex`.

Measure	Value
Rows	5,679
Analyzable sign tokens	18,044
Unique analyzable codes	712
Eroded 000 tokens	1,881
Encoded-space 999 tokens	17
Dominant analyzable codes, ≥ 500 tokens	6
Hapax codes	218
Codes occurring one to five times	448

Table 10: Code-level sign inventory overview.

8.3 Frequency Compression

The frequency profile is highly compressed. Six dominant analyzable codes account for 4,970 tokens; 43 codes with at least 100 occurrences account for 12,151 tokens. Conversely, 448 codes occur five times or fewer. These rare codes are the first candidates for allograph checks, damaged readings, narrow regional variants, and possible cataloging noise.

Band	Codes	Tokens
Dominant, ≥ 500	6	4,970
Common, 100–499	37	7,181
Active, 20–99	71	3,421
Low, 6–19	150	1,573
Rare, 3–5	125	471
Dis legomena	105	210
Hapax	218	218

Table 11: Frequency bands for analyzable sign codes.

Coverage	Codes required	Tokens covered	Total tokens
0.50	21	9,027	18,044
0.75	61	13,558	18,044
0.90	158	16,248	18,044
0.95	264	17,145	18,044
0.99	532	17,864	18,044

Table 12: Number of most frequent codes needed to cover the analyzable sign-token stream.

8.4 Candidate Functional Classes

The first functional labels are deliberately mechanical. A code is labeled as an initial or terminal candidate only if it has at least 20 positional observations and reaches a 0.75 edge-position threshold. A medial candidate must have at least 20 positional observations and reach a 0.85 medial-position threshold.

Class	Leading codes	Criterion	Interpretive status
Initial	501, 817, 034, 692, 503	Start proportion ≥ 0.75	Behavioral, not phonetic
Terminal	400, 527, 151, 520, 700, 154, 090	End proportion ≥ 0.75	Behavioral, not grammatical
Medial	845, 752, 060, 760, 585, 500, 100, 002	Medial proportion ≥ 0.85	Robust internal-formula target

Table 13: First code-level functional classes derived from positional behavior.

8.5 Compression Policy

The project will use four compression layers:

1. **Frozen code layer:** preserve every local code exactly as recorded.
2. **Functional layer:** mark dominant, initial, terminal, medial, singleton, rare, and eroded categories.
3. **Variant-candidate layer:** identify rare codes and neighboring numerical families for image and concordance review.
4. **Reduced-grapheme layer:** merge only after a Wells/ICIT crosswalk, image inspection, and distributional-neighbor comparison agree.

No code will be merged solely because it is rare. Rarity is a triage signal, not an argument.

9 Crosswalk and Neighbor-Profile Clustering

9.1 Purpose

This milestone creates two linked research layers. First, it defines a provisional crosswalk from local sign code to ICIT/Wells code. Second, it groups signs by distributional neighbor behavior so that possible allographs or functional equivalents can be prioritized for visual and catalog review.

The ICIT documentation explicitly describes sign grouping by similar preceding or following sign-neighbor frequencies and treats this as a way to identify similar paradigmatic behavior [Fuls, 2026]. The present implementation follows the same spirit but keeps the method simple and reproducible: each sign is represented by a vector of preceding and following neighbors in normalized internal reading order, and sign pairs are scored by cosine similarity.

9.2 Provisional Crosswalk

The crosswalk is deliberately conservative. Local codes are mapped to identical ICIT/Wells codes only as a provisional identity relation, because the corpus itself uses ICIT-style numeric sign-code notation. Two non-sign codes are separated from the sign layer:

- 000: eroded or illegible sign;
- 999: encoded blank space between signs.

Crosswalk status	Codes	Tokens
Provisional identity	712	18,044
Eroded unknown	1	1,881
Encoded space	1	17

Table 14: Provisional local-code to ICIT/Wells crosswalk status.

9.3 Neighbor Similarity

The script `scripts/crosswalk-neighbor-analysis.ps1` used 3,517 texts for neighbor profiles and skipped 2,162 texts because they were incomplete, damaged, directionally unsuitable, or contained eroded signs. It considered 294 eligible sign codes with at least five observed tokens and generated 25,193 pairwise similarity rows.

Similarity is not an allograph claim. It is a review priority. A high score means that two signs tend to occur in similar left and right environments; only image inspection, sign-catalog comparison, and artifact-context checks can determine whether they are graphic variants, functional substitutes, or merely distributional companions.

Sign A	Sign B	Tokens A	Tokens B	Cosine
027	028	6	5	1.000000
636	642	22	8	0.999777
347	849	10	8	0.992278
110	777	6	7	0.991241
027	905	6	13	0.990443
526	527	13	63	0.982841
817	861	221	272	0.965020
817	820	221	244	0.956926
705	706	219	102	0.953396

Table 15: Selected high-similarity neighbor-profile pairs.

9.4 Review Clusters

At a cosine threshold of 0.90, the first pass yields 18 review clusters. Several clusters are immediately interesting because they reproduce or approximate groupings discussed in ICIT-style positional work: for example, 817/820/861 form a strong initial-sign neighborhood, and 705/706 form a tight pair. Other clusters may reflect terminal behavior, numeral/marker behavior, or sparse-data artifacts.

Cluster	Signs	Token sum
N001	176 222 362 482 630 760 772 773 792 923 927	542
N002	090 151 161 400 520 565 621 679 740	3,109
N005	817 820 861 880	748
N014	705 706	321
N016	033 034	707

Table 16: Selected neighbor-profile review clusters.

9.5 Next Action

The next step is image/catalog review for the highest-value pairs and clusters. Priority should go to pairs that combine high similarity with enough tokens to be reliable, such as 817/861, 817/820, 705/706, 033/034, and 526/527. Sparse pairs remain useful but should be treated as low-confidence prompts for inspection rather than analytic conclusions.

10 Visual and Catalog Review

10.1 Purpose

The neighbor-profile analysis nominates signs that behave similarly. It does not show that the signs look alike. This milestone therefore creates a visual and catalog review pack: a prioritized list of candidate pairs, same-text examples, source-access status, and a protocol for deciding whether a pair should be merged, linked as variants, kept distinct, or held pending.

The method follows two constraints from prior scholarship. First, ICIT-style grouping uses similar sign-neighbor frequencies to identify paradigmatic behavior [Fuls, 2026]. Second, allograph claims require image and context review: mirroring, writing direction, space-saving, regional anomalies, carving errors, and catalog errors must be checked before reducing the sign list [Daggumati and Revesz, 2021]. The Harappa review of Fuls’ catalog emphasizes the catalog’s visual documentation and sign-reference function [Harappa.com, 2026], but the present workspace does not yet contain the full sign-image coverage needed for a final verdict.

10.2 Source Status

10.3 Partial Visual Evidence Found

The local Fuls 2022 corpus sample is a major upgrade for visual review. PDF page 21 lists 205 high-number ICIT signs and contains 205 embedded sign images; layout-ordered extraction gives direct catalog visuals for 705, 706, 817, 820, 861, and 920. The Fuls 2013 positional-analysis article supplies the missing 692 panel on PDF page 10. This gives both visual sides for all four Tier 1 pairs. These are visual-review inputs only; no merge or variant claim is licensed until artifact-level checks are complete.

10.4 Review Tiers

The script `scripts/visual-catalog-review.ps1` ranks the top 80 distributional pairs into review tiers. The ranking favors high cosine similarity but penalizes sparse evidence. All rows remain marked as *Pending* for visual/catalog status.

10.5 Priority Pairs

10.6 Review Protocol

Each pair must pass through six questions:

1. **Shape:** Are the signs graphically similar enough to be variants?
2. **Direction:** Could mirroring or reversal explain the difference?
3. **Space:** Could crowding or line-end compression explain the difference?
4. **Context:** Do positions, neighbors, sites, and artifact types support functional equivalence?
5. **Independence:** Do the occurrences reflect independent texts rather than a repeated-object effect?

Source	Status	Use
Committed workspace image corpus	Unavailable	No committed sign or artifact image assets are present; scratch extracts remain local.
Fuls 2013 positional analysis	Tracked local PDF	Supplies the page 10 692 panel and positional-method argument.
Fuls 2022 <i>Corpus of Indus Inscriptions</i>	Tracked local PDF	Searchable 35-page corpus sample; page 21 gives a high-number ICIT sign-list panel.
Fuls 2023 SearchInside local PDF	Tracked supplementary sample	Searchable 35-page catalog sample with embedded sign-list images; confirms additional visuals including 820 and 920.
ICIT/Epigraphica	Catalog described, images not harvested	Primary catalog target.
Deutsche Digitale Bibliothek	Bibliographic record only	Confirms Fuls 2023 catalog; contents only openly accessible.
Harappa catalog review	Public descriptive source	Confirms catalog scope and importance.
Daggumati and Revesz 2021	Open-access method source	Supplies allograph-review checks.
CISI Vol. 1 local PDF	Tracked local PDF	Artifact-image authority for later seal and inscription checks.
Marshall 1931 Mohenjo-daro Vol. II	Tracked local PDF	Archaeological context and early excavation reference.
CISI remaining volumes	Not locally available	Additional artifact-image authorities to seek later if needed.

Table 17: Current source status for visual and catalog review.

Sign	PDF page	Related pair(s)	Status
692	10	692/920	Pair-side visual evidence
705	21	705/706	Pair-side visual evidence
706	21	705/706	Pair-side visual evidence
817	21	817/861; 817/820	Pair-side visual evidence
820	21	817/820; 820/861	Pair-side visual evidence
861	21	817/861; 820/861	Pair-side visual evidence
920	21	692/920	Pair-side visual evidence

Table 18: Partial visual evidence located in the local Fuls corpus and catalog PDF samples.

Tier	Pairs
Tier 1	4
Tier 2	3
Tier 3	26
Sparse-data hold	47

Table 19: Review tiers in the first visual/catalog candidate pack.

Sign A	Sign B	Cosine	Tokens A	Tokens B	Tier
817	861	0.965020	221	272	1
817	820	0.956926	221	244	1
705	706	0.953396	219	102	1
692	920	0.953170	77	153	1
526	527	0.982841	13	63	2
090	565	0.953096	195	15	2
400	565	0.950984	502	15	2
820	861	0.924203	244	272	3
033	034	0.908290	523	184	3

Table 20: Selected visual/catalog review candidates.

6. **Verdict:** Merge, variant-link, keep distinct, or hold pending.

10.7 Outputs

- `outputs/visual_catalog_review_candidates.csv`
- `outputs/visual_catalog_review_examples.csv`
- `outputs/visual_catalog_review_protocol.csv`
- `outputs/visual_catalog_source_status.csv`
- `outputs/visual_catalog_found_signs.csv`
- `outputs/visual_catalog_review.tex`

10.8 Interpretation

The first review pack identifies 817/861, 817/820, 705/706, and 692/920 as the strongest immediate candidates for visual/catalog inspection. The local PDF library now supplies both visual sides for all four pairs. No sign merge is proposed at this stage.

11 Tier 1 Visual Adjudication

11.1 Purpose

With 692 now located in the local Fuls 2013 positional-analysis article, the project has both visual sides for all four Tier 1 distributional candidates. This section converts evidence collection into a first visual adjudication. The goal is not to merge signs. The goal is to separate plausible allograph candidates from signs that merely share a positional or functional profile.

The adjudication follows the visual-caution protocol: catalog shape, positional behavior, artifact context, direction, and object-level independence must all support a merge before the inventory is changed [Daggumati and Revesz, 2021]. Fuls’ positional analysis is especially useful here because it argues that visually similar signs can still be separate graphemes when their positional behavior differs [Fuls, 2013].

Pair	Visual assessment	Provisional decision
817/861	Moderate family resemblance: both are enclosed signs with angular internal marking, but the outline and inner geometry differ.	Keep distinct. Treat as a functional-cluster candidate, not a merge candidate.
817/820	Moderate family resemblance: both are enclosed signs, but 817 has a simpler chevron-like interior while 820 has a divided/crossed interior.	Keep distinct. Treat as a functional-cluster candidate, not a merge candidate.
705/706	Strong visual resemblance: same open-U/long-stem family, with 706 adding or emphasizing an inner nested stroke.	Variant-link candidate. Highest priority for artifact-level review; no merge yet.
692/920	Weak visual resemblance: 692 is crossed/X-shaped, while 920 is a curved single-stroke form.	Reject visual-allograph hypothesis at catalog level. Keep as a functional-cluster/control case.

Table 21: First visual adjudication of Tier 1 distributional candidates.

11.2 Provisional Decisions

11.3 Interpretation

The important result is that distributional similarity is not the same as visual equivalence. The strongest allograph-style candidate is 705/706. The pairs 817/861 and 817/820 look more like members of a shared initial-sign family than variants of one sign. The pair 692/920 is now valuable as a control case: it is distributionally strong but visually weak.

11.4 Outputs

- `outputs/tier1_visual_adjudication.csv`
- `outputs/tier1_visual_adjudication.tex`
- `outputs/tier1_artifact_review_queue_705_706.csv`
- `outputs/tier1_artifact_review_summary_705_706.csv`
- `outputs/tier1_artifact_review_705_706.tex`

12 Sign-Family Hypothesis Model

12.1 Motivation

The visual-review stage changes the question from “are these signs identical?” to “what kind of difference are we seeing?” A pair of signs may be distinct signs, regional allographs, chronological variants, base-plus-modifier forms, members of a shared syntactic family, or visually similar but functionally unrelated shapes. This section begins a computational model for those alternatives.

The model is inspired by the observation that later South Asian scripts preserve related graphic habits across regional scripts, but it does not assume direct descent from the Indus system to Brahmi or any Brahmic descendant. The Brahmic comparison is used only as a

hypothesis generator: if base signs plus added strokes, doubling, mirroring, or other modifications mattered in the Indus corpus, they should leave measurable traces in position, neighbors, region, chronology, object type, or text class.

12.2 Features

The script `scripts/sign-family-hypothesis-model.ps1` takes the visual-review candidates and scores them with auditable features:

- neighbor-profile cosine similarity;
- positional similarity from initial, medial, and terminal proportions;
- Jensen-Shannon divergence across region, site, object type, period, phase, and time metadata;
- permutation simulations for distributional separation using a fixed random seed;
- visual priors only for catalog-reviewed Tier 1 pairs.

The scores are heuristic. They rank hypotheses for review; they do not change the sign inventory.

12.3 Hypothesis Classes

Modifier or variant candidate visually close signs where an added stroke or related graphic operation may carry value.

Visual functional family visually related signs that should remain distinct unless artifact-level review supports a stronger claim.

Functional cluster, not allograph signs that behave alike distributionally but do not look alike.

Regional allography test signs whose similarity coexists with unusually separated site or regional distributions.

Chronological layering test signs whose similarity coexists with period, phase, or time separation; this remains provisional because chronological metadata are sparse.

12.4 Current Tier 1 Result

Pair	Model hypothesis	Allograph	Regional	Chron.	Modifier
705/706	Modifier or variant candidate	0.936	0.044	0.043	0.741
817/820	Visual functional family	0.850	0.053	0.049	0.489
817/861	Visual functional family	0.819	0.021	0.033	0.514
692/920	Functional cluster, not allograph	0.705	0.044	0.042	0.116

Table 22: First sign-family hypothesis scores for Tier 1 visual candidates.

12.5 Interpretation

The model supports the current research direction. 705/706 is the strongest base-plus-modifier or variant-link candidate. 817/820 and 817/861 look like a related initial-sign family rather than a simple merge. 692/920 is valuable as a control case: high distributional similarity does not imply visual equivalence.

The broader output also creates new queues for regional and chronological tests. These should be treated as computational leads, not conclusions, because chronology fields are incomplete and many pairs still lack visual adjudication.

12.6 Outputs

- `outputs/sign_family_pair_features.csv`
- `outputs/sign_family_hypothesis_scores.csv`
- `outputs/sign_family_model_summary.csv`
- `outputs/sign_family_model.tex`

13 Core Inventory Model

13.1 Motivation

The corpus has hundreds of sign codes, but most tokens are carried by a much smaller core. This matters for the working hypothesis that the system may contain a compact phonetic or morphemic inventory plus extensions: modifiers, ligatures, regional variants, chronological variants, numbers, rare symbols, object marks, or catalog over-splitting. The comparison to Sanskrit or later Brahmic scripts is used only as a model-shape analogy. It is not a claim that the Indus language was Sanskrit or that the script is directly Brahmi.

13.2 Method

The script `scripts/core-inventory-model.ps1` excludes 000 and 999, orders signs by frequency, assigns coverage groups, and computes:

- cumulative token coverage;
- normalized positional entropy across ten text-position bins;
- left- and right-neighbor entropy;
- dominant left and right neighbor shares;
- region, site, and object-type entropy;
- a first mechanism hypothesis for each sign.

13.3 Coverage Result

The practical result is that 60 signs cover about 75 percent of analyzable tokens, and 157 signs cover about 90 percent. This supports a fast modeling strategy: build the grammar of the core and extension layer first; treat the long tail as a later classification problem.

Group	Signs	Token pct.	Dominant role	Dominant mechanism
Core50	20	0.488	Medial carrier	Core phonetic/morphemic candidate
Core75	40	0.260	Flexible carrier	Core phonetic/morphemic candidate
Extension90	97	0.152	Medial carrier	Extended inventory candidate
Extension95	106	0.050	Mixed position	Formula or grammar marker
LongTail99	268	0.040	Mixed position	Bound modifier/formula variant candidate
RareTail	181	0.010	Mixed position	Bound modifier/formula variant candidate

Table 23: Core inventory coverage groups after excluding eroded and blank codes.

Test	Value	Interpretation
Signs covering 75 percent of analyzable tokens	60	Compatible with a compact phonetic core plus extensions, but not diagnostic by itself.
Core75 flexible/medial/mixed carriers	43	Supports testing a phonetic or morphemic carrier layer.
Core75 initial or terminal markers	17	Warns that formula grammar is substantial; this is not a plain alphabet model.
Signs covering 90 percent of analyzable tokens	157	Practical all-sign modeling target before sparse-tail uncertainty dominates.
Signs after 95 percent coverage	449	Long tail may include variants, numbers, names, object marks, rare logograms, or catalog over-splitting.
Tail signs with dominant left/right neighbor	188	Leads for diacritic, conjunct, affix, or formula-variant behavior.

Table 24: Compatibility tests for a compact-core-plus-extension model.

13.4 Abugida-Scale Compatibility Tests

13.5 Interpretation

The working model is now faster and sharper: do not attempt to decipher 700 or 900 signs equally. First model the 60-sign core and the 157-sign 90-percent layer. Within that layer, distinguish carrier signs from formula or grammar markers. Then test whether the long tail behaves like modifiers, conjuncts, numerals, regional spellings, chronological layers, rare logograms, or noise from catalog over-splitting.

This result is compatible with an abugida-like system in the broad structural sense: compact high-frequency core plus extended graphic inventory. It is also compatible with a mixed administrative writing system. The next tests must therefore discriminate between those possibilities rather than choose the more exciting one too early.

13.6 Outputs

- `outputs/core_inventory_sign_model.csv`
- `outputs/core_inventory_groups.csv`
- `outputs/core_inventory_abugida_tests.csv`
- `outputs/core_inventory_model.tex`

14 Onomastic Anchor Model

14.1 Motivation

The Linear-Elamite-style proper-name strategy is useful as a search pattern: if a writing system contains royal names, personal names, titles, lineage labels, toponyms, or office markers, those elements may recur in formulaic positions and create the first phonetic footholds. The Indus corpus does not give us known kings or bilingual readings, so the first version of this strategy must remain internal. We search for name-like anchors without assigning values.

14.2 Method

The script `scripts/onomastic-anchor-model.py` enumerates one- to five-sign n-grams in normalized reading order, excluding 000 and 999. For each repeated n-gram it records frequency, text count, site and region spread, type spread, position, dominant left and right neighbors, immediate frame, seal share, and complete-text share.

The model then gives each n-gram four auditable scores:

- **Name anchor score:** repeated two- to five-sign blocks with stable context and complete/seal support.
- **Title marker score:** short high-frequency signs or bigrams with fixed position and diverse neighboring material.
- **Cross-site anchor score:** name-like blocks whose distribution is not restricted to a single findspot.
- **Formula risk score:** a caution score for blocks that may be administrative formula rather than names.

The same script also scores immediate frames of the form `left _ right`. A high-scoring frame has many different fillers inside the same local environment, which is exactly the kind of slot where names or titles might hide.

14.3 Corpus-Wide Result

Metric	Value
Analyzable texts	5531
Complete analyzable texts	3673
Repeated n-grams	3356
Strong title/classifier candidates	17
Strong repeated name-like candidates	299
Cross-site name-anchor candidates	35
Variable formula-slot candidates	1013

Table 25: Onomastic-anchor search results using repeated one- to five-sign n-grams.

14.4 Title and Classifier Leads

Sign/ngram	Count	Sites	Start pct.	End pct.
520	321	16	0.044	0.822
820	244	18	0.775	0.164
151	94	10	0.075	0.883
817	221	14	0.873	0.077
003	259	14	0.583	0.108
390	290	25	0.128	0.514
740	1928	41	0.093	0.700
235	250	12	0.432	0.064

Table 26: Top title/classifier-marker leads. These signs are not assigned meanings; the table only says that their distribution looks formulaic and positionally marked.

Several of these leads agree with earlier positional results. Signs 520, 151, and 740 are strongly terminal; 820, 817, and 003 are strongly initial. This split is important: an onomastic model should not treat every repeated sign as a name. Some signs probably mark a role, object class, title, suffix, or formula boundary.

14.5 Name-Like Block Leads

N-gram	Count	Sites	Name score	Formula risk
002-003-390	14	5	0.966	0.572
706-033-520	16	3	0.943	0.598
017-585-740	14	4	0.927	0.593
016-390	14	5	0.924	0.612
920-060-741	38	7	0.923	0.640
820-060	26	5	0.934	0.620

Table 27: High-ranking repeated blocks for artifact-level review. These are candidates for names, offices, lineages, places, or fixed administrative phrases.

The strongest immediate action is not phonetic reading. It is artifact-level review. For each candidate, we need to inspect whether the block occurs on seals, tablets, pottery, or mixed

media; whether the surrounding signs are stable; whether variants replace one another in the same slot; and whether the visual artifact context suggests owner, authority, commodity, place, or ritual formula.

14.6 Variable Formula Slots

Left	Right	Count	Fillers	Slot score
002	740	345	247	0.957
060	<END>	193	176	0.949
<START>	390	236	163	0.941
002	<END>	705	508	0.937
741	<END>	198	163	0.936
060	740	101	85	0.928

Table 28: Top variable formula frames. A high filler count means the same local environment admits many different inner blocks, making it a plausible grammatical or onomastic slot.

The frame 002 _ 740 is especially important: it has 345 occurrences and 247 distinct fillers. This looks less like a single word and more like a productive slot inside a formula. If names exist in the corpus, this kind of environment is one of the best places to search.

14.7 Interpretation

This stage gives the project its first proper-name hunting protocol. The model does not assume kings, Sanskrit, Dravidian, Brahmi, or any particular phonetic value. It identifies slots and repeated blocks where names and titles would be likely if the corpus contains them. The next milestone is an artifact review queue for the strongest title markers, cross-site anchors, and variable slots, beginning with candidates that overlap our visually reviewed signs such as 705/706, 817/820/861, and 692/920.

14.8 Outputs

- `outputs/onomastic_anchor_ngram_candidates.csv`
- `outputs/onomastic_title_marker_candidates.csv`
- `outputs/onomastic_formula_slots.csv`
- `outputs/onomastic_anchor_summary.csv`
- `outputs/onomastic_anchor_model.tex`

15 Proto-Decipherment Scaffold

15.1 Purpose

The project can now move from distributional description toward controlled decipherment. The goal of this stage is not to pronounce signs phonetically. It is to establish a pre-phonetic scaffold: which signs behave like title/classifier markers, which frames behave like productive slots, which repeated blocks deserve artifact-level review, and which abstract inscription templates recur across the corpus.

This is the practical bridge between statistics and decipherment. A future reading must satisfy these constraints before it can be treated as more than an inspired guess.

15.2 Method

The script `scripts/proto-decipherment-scaffold.py` consumes the onomastic-anchor outputs and the cleaned corpus. It selects:

- title/classifier markers with title-marker score at least 0.78;
- name-like blocks with name-anchor score at least 0.90 and formula-risk score at most 0.66;
- productive slots with slot score at least 0.91.

It then templates every analyzable inscription in normalized reading order. Tokens can receive abstract roles such as `INITIAL_TITLE_OR_CLASSIFIER`, `TERMINAL_TITLE_OR_SUFFIX`, `SLOT_FILLER`, or `NAME_LIKE_BLOCK`. These are deliberately functional labels, not translations.

15.3 Corpus-Wide Result

Metric	Value
Title/classifier markers used	17
Name-like blocks used	52
Productive slots used	20
Texts templated	5531
Texts with any anchor	3621
Template families	608
Artifact review rows	1287
Explicit constraint rows	53

Table 29: Proto-decipherment scaffold summary.

15.4 Concrete Functional Leads

Candidate	Functional lead	Count	Start pct.	End pct.
520	Terminal title/suffix	321	0.044	0.822
820	Initial title/classifier	244	0.775	0.164
151	Terminal title/suffix	94	0.075	0.883
817	Initial title/classifier	221	0.873	0.077
003	Leading formula marker	259	0.583	0.108
390	Closing formula marker	290	0.128	0.514
740	Terminal title/suffix	1928	0.093	0.700
861	Initial title/classifier	272	0.654	0.191
400	Terminal title/suffix	502	0.048	0.912

Table 30: High-confidence functional leads. The label is a role constraint, not a semantic translation.

These signs give the first concrete grammar-like skeleton. Signs 820, 817, and 861 repeatedly appear at the beginning of inscriptions. Signs 520, 151, 400, and 740 repeatedly appear at the end. Any proposed reading that treats these signs as ordinary interchangeable phonetic letters must explain this positional behavior.

Frame	Occurrences	Distinct fillers
002 _ 740	345	247
060 _ END	193	176
START _ 390	236	163
002 _ END	705	508
741 _ END	198	163
060 _ 740	101	85
820 _ END	170	147

Table 31: Productive slots with many different fillers in a stable immediate frame.

15.5 Productive Slots

The frame 002 _ 740 is now one of the project’s most important decipherment targets. It is productive enough that the filler is unlikely to be random ornament. If the script records names, offices, places, commodities, or grammatical phrases, this frame is a prime hunting ground.

15.6 First Abstract Template Families

Template	Count	Sites
SIGN SIGN	827	29
SIGN	773	31
SIGN TERMINAL_TITLE_OR_SUFFIX	345	12
SIGN SIGN TERMINAL_TITLE_OR_SUFFIX	266	20
TERMINAL_TITLE_OR_SUFFIX	164	14
NAME_LIKE_BLOCK[3] TERMINAL_TITLE_OR_SUFFIX	58	3
LEADING_FORMULA_MARKER	57	7
TERMINAL_TITLE_OR_SUFFIX		

Table 32: Top abstract template families after collapsing selected anchor blocks.

The important point is not that the most common templates are already deciphered. The important point is that the corpus can now be sorted into recurrent functional shapes. This lets us test readings across template families rather than one inscription at a time.

15.7 Concrete Constraints for Future Readings

The file `outputs/proto_decipherment_constraints.csv` records explicit rules that future decipherment attempts must satisfy. Examples:

- A reading of 520 must explain why it is terminal in 82.2 percent of occurrences.
- A reading of 820 must explain why it is initial in 77.5 percent of occurrences.
- A reading of 740 must explain why it is extremely common and terminal-heavy.
- A reading of 002 _ 740 must explain why the frame has 345 occurrences but 247 distinct fillers.
- A reading of a repeated block such as 820-060 must survive every occurrence in the review queue, not merely one attractive artifact.

15.8 Interpretation

This is the first stage where the project can responsibly talk about a proto-decipherment. The current result suggests a formulaic system with identifiable initial markers, terminal markers, and productive internal slots. That is compatible with owner names, titles, offices, places, commodities, lineage labels, grammatical endings, or some mixture of these. It is not yet a phonetic decipherment.

The next hard push should pair this scaffold with images from the CISI/Marshall corpus. If a productive slot repeatedly occurs on seals with similar iconography or artifact function, we can begin testing whether the slot encodes owner, authority, commodity, place, or title. This is also where external onomastic evidence from Mesopotamian references to Meluhha should eventually be brought in, but only as a test set, not as a source of free guesses.

15.9 Outputs

- `outputs/proto_decipherment_text_templates.csv`
- `outputs/proto_decipherment_template_families.csv`
- `outputs/proto_decipherment_anchor_review_queue.csv`
- `outputs/proto_decipherment_constraints.csv`
- `outputs/proto_decipherment_summary.csv`
- `outputs/proto_decipherment_scaffold.tex`

16 Slot Paradigm Model

16.1 Purpose

The proto-decipherment scaffold identified productive frames such as 002 _ 740. This section asks the next, harder question: do the fillers inside those frames behave like paradigms? If a slot contains many fillers, and those fillers show recurring final signs, initial signs, cross-frame reuse, or one-edit alternations, then the system is beginning to expose grammar-like structure.

This is the first place where we can responsibly write proto-glosses. A proto-gloss is a functional label such as `TERMINAL MARKER`, `SLOT INTRODUCER`, `FINAL MODIFIER`, or `NAME/TITLE SLOT`. It is not a phonetic value and not a translation.

16.2 Method

The script `scripts/slot-paradigm-model.py` extracts fillers up to five signs long from the highest-scoring productive frames. For each frame and filler it records artifact metadata, site spread, object type, first sign, final sign, stem-minus-final, and examples. It then computes:

- frame-level filler diversity;
- high-reuse final signs inside slot fillers;
- high-reuse initial signs inside slot fillers;
- stems that occur with multiple final signs;
- fillers reused across multiple frames;
- minimal one-edit alternations within the same frame.

16.3 Corpus-Wide Result

Metric	Value
Productive frames modeled	20
Slot occurrences extracted	4279
Distinct frame/filler pairs	3060
Final-sign leads	335
Initial-sign leads	377
Alternating stem/final paradigms	100
Cross-frame reused fillers	189
Minimal filler pairs	1043

Table 33: Slot-paradigm extraction results.

16.4 Productive Frames

Frame	Occurrences	Fillers	Mean length	Top filler
002 _ 740	345	247	2.267	048
060 _ END	193	176	2.985	690-740
START _ 390	236	163	2.411	590
002 _ END	705	508	2.766	240-520
741 _ END	198	163	3.005	001-055-220-740-090
060 _ 740	101	85	2.248	690
820 _ END	170	147	3.041	002

Table 34: Highest-priority productive frames.

The frame 002 _ 740 remains the fastest route toward concrete decipherment. It is frequent, has 247 distinct fillers, and the fillers average just over two signs. This is compatible with a compact name/title/place/commodity slot rather than a long sentence-like field.

16.5 First Proto-Gloss Constraints

The following functional labels are now strong enough to use as working constraints:

- 740: terminal-heavy marker and high-reuse final sign inside slots. Proto-gloss: **terminal/formula final**.
- 520: terminal-heavy marker and high-reuse final sign. Proto-gloss: **terminal title/suffix candidate**.
- 820, 817, 861: initial-heavy markers. Proto-gloss: **initial title/classifier candidates**.
- 002: frequent initial element inside productive slot fillers and the left boundary of major frames. Proto-gloss: **slot introducer or internal classifier candidate**.
- 002 X 740: productive formula. Proto-gloss: **X in a name/title/place/commodity slot**, with X still undeciphered.

These are not readings in the phonetic sense. They are constraints that any future phonetic reading must satisfy.

16.6 Final-Sign Leads

Final sign	Count	Stems	Frames	Role lead
740	870	512	14	High-reuse final modifier
520	242	114	13	High-reuse final modifier
220	133	69	15	High-reuse final modifier
390	151	67	16	High-reuse final modifier
090	118	56	12	High-reuse final modifier
760	59	43	9	High-reuse final modifier
590	78	42	8	High-reuse final modifier
400	70	42	12	High-reuse final modifier
100	57	39	9	High-reuse final modifier

Table 35: Final signs that recur after many different stems inside productive slots.

This is one of the strongest grammar signals so far. Signs such as 740, 520, 220, 390, and 090 are not merely frequent; they close many different fillers across many frames. That is compatible with suffixes, classifiers, titles, formula closers, or final modifiers.

16.7 Initial-Sign Leads

Initial sign	Count	Remainders	Frames	Role lead
002	444	306	13	High-reuse initial modifier
240	255	110	15	High-reuse initial modifier
032	162	96	18	High-reuse initial modifier
235	125	75	14	High-reuse initial modifier
220	119	75	16	High-reuse initial modifier
741	73	66	9	High-reuse initial modifier
031	88	62	16	High-reuse initial modifier
233	84	58	10	High-reuse initial modifier
820	84	49	10	High-reuse initial modifier

Table 36: Initial signs that introduce many different remainders inside productive slots.

Initial signs such as 002, 240, 032, and 235 may be operating as classifiers, prefixes, stem initials, or formula-internal introducers. The critical point is that they govern many different following remainders and therefore should not be read only as isolated icons.

16.8 Paradigms and Alternations

The script found 100 cases where the same stem occurs with multiple final signs. Examples include:

- in frame 002 _ 740, stem 240 occurs with twelve different final signs;
- in frame 032 _ 740, stem 220 occurs with ten different final signs;
- in frame START _ 390, stem 861-002 occurs with eight different final signs.

The script also found 189 fillers reused across multiple frames. Some are single signs, but longer forms such as 176-740, 590-390-740, 240-520, and 705-033-520 are more interesting because they can be tracked as possible reusable formula pieces or name/title blocks.

16.9 Interpretation

The stones are not speaking phonetically yet, but they are beginning to speak grammatically. The present evidence supports a working model with:

- initial classifier/title-like signs;
- terminal final/title/suffix-like signs;
- productive internal slots;
- repeated slot fillers;
- stems that take alternate final signs.

The next fastest path is to pair the paradigm tables with seal images and iconography. If a filler family such as 002 X 740 correlates with a symbol, object type, or site pattern, then X can be tested as owner, office, lineage, place, commodity, or formulaic title. That is the bridge from functional proto-glosses toward decipherment.

16.10 Outputs

- `outputs/slot_paradigm_summary.csv`
- `outputs/slot_paradigm_frames.csv`
- `outputs/slot_paradigm_fillers.csv`
- `outputs/slot_paradigm_occurrences.csv`
- `outputs/slot_paradigm_final_signs.csv`
- `outputs/slot_paradigm_initial_signs.csv`
- `outputs/slot_paradigm_stem_finals.csv`
- `outputs/slot_paradigm_cross_frame_reuse.csv`
- `outputs/slot_paradigm_minimal_pairs.csv`
- `outputs/slot_paradigm_model.tex`

17 Structural Reconstruction Model

17.1 Purpose

The project can now begin reconstructing sentence-like structure. Because most Indus inscriptions are short, the safer term is formula reconstruction rather than full sentence translation. The goal is to assign functional roles to signs and slots, propose competing semantic frames, and identify the rows where phonetic testing can begin.

This section remains pre-phonetic. A structural reconstruction such as INITIAL CLASSIFIER + 002 X 740 + TERMINAL is not a translation. It is a constrained parse that future semantic and phonetic hypotheses must satisfy.

17.2 Method

The script `scripts/structural-reconstruction-model.py` combines the previous models:

- title/classifier candidates from the onomastic model;
- productive slots and fillers from the slot-paradigm model;
- name-like n-grams from the onomastic anchor model;
- high-reuse initial and final signs from the paradigm model;
- cross-frame filler reuse and minimal-pair evidence for phonetic readiness.

For each analyzable inscription, the script emits a structural parse, a compressed structural template, a semantic frame, semantic alternatives, and a phonetic-readiness score. Productive slots are given priority over broader name-like blocks so that a long attractive n-gram cannot hide a more informative internal paradigm.

17.3 Corpus-Wide Result

Metric	Value
Texts reconstructed	5531
Structural templates	563
Semantic frame families	8
Medium/high phonetic-readiness texts	607
Functional sign proto-glosses	32
Productive frames used	20
Name-like blocks used	52
Cross-frame filler leads	41

Table 37: Structural reconstruction summary.

17.4 Semantic Frame Families

Semantic frame	Count
LowEvidenceSequence	2534
TerminalFormula	1892
InitialClassifierFormula	368
PrimeEntitySealFormula	279
NameLikeTerminalFormula	238
ClassifierSlotTerminalFormula	156
PrimeEntityAdministrativeFormula	61
ClassifierTerminalFormula	3

Table 38: High-level semantic frame assignments. These are competing frame hypotheses, not translations.

The strongest semantic target is `PrimeEntitySealFormula`. These are mostly seal inscriptions containing the productive 002 X 740 frame. The current semantic alternatives for `X` are owner/name, office/title, lineage, place, commodity, or authority formula.

17.5 Functional Proto-Glosses

Sign	Functional proto-gloss
740	Terminal title/suffix; slot final modifier
520	Terminal title/suffix; slot final modifier
400	Terminal title/suffix; slot final modifier
151	Terminal title/suffix; slot final modifier
820	Initial title/classifier
817	Initial title/classifier
861	Initial title/classifier
002	Slot initial/final modifier; major slot boundary partner
390	Closing formula marker; slot final modifier
235	Leading formula marker; slot initial/final modifier
240	Slot initial modifier
032	Slot initial modifier

Table 39: Functional proto-glosses now strong enough to guide reconstruction. No phonetic value is assigned.

17.6 Example Structural Reconstructions

The following high-confidence examples show inscription-level formula reconstruction. Bracketed material is the slot filler.

- **L-41, Lothal** (PrimeEntitySealFormula):
START_233[817-002] + 002_740[233] + 233_END[740-090]
- **M-708, Mohenjo-daro** (PrimeEntitySealFormula):
START_390[140] + 390 + 002_740[061] + 740
- **M-246, Mohenjo-daro** (PrimeEntitySealFormula):
START_240[817-002] + 002_740[240-415-100] + 740
- **C-53, Chanhudaro** (PrimeEntitySealFormula):
861 + 002_740[235-222] + 740 + 031_END[032]
- **K-16, Kalibangan** (ClassifierSlotTerminalFormula):
START_240[820-060] + 060_END[240-740-090]

The important advance is that the model can now reconstruct inscriptions as linked formulae, not merely as sign strings. For example, 817-002-233-740-090 is not just five signs. It can be parsed as an initial frame, a prime 002 X 740 entity slot, and a terminal frame.

17.7 Phonetic Bootstrapping

The file `outputs/phonetic_bootstrap_candidates.csv` contains 120 prioritized rows for the next stage. A row becomes phonetic-ready when it combines several kinds of evidence:

- complete or seal context;
- productive slot membership;
- cross-frame reused filler;

- minimal-pair or one-edit alternation evidence;
- clear structural parse.

The next phonetic move should not be a global assignment like “sign 740 equals sound X”. It should be a controlled test: pick a reused filler, inspect all artifacts where it appears, compare its frames, and ask whether a proposed sound or morpheme predicts every occurrence. This is especially promising for fillers reused across 002_740, 060_END, START_240, and related productive frames.

17.8 Interpretation

We can now say that the project has reached structural proto-decipherment. The inscriptions are not translated, but many can be parsed into functional formulae with semantic alternatives. The best current working model is:

CLASSIFIER/TITLE + SLOT INTRODUCER + ENTITY FILLER + TERMINAL/FINAL

The semantic identity of the entity filler remains open: personal name, title, office, lineage, place, commodity, or administrative phrase. The fastest route forward is to connect these reconstructions to the seal images and iconography. If a structural frame correlates with a symbol, object type, or site-specific artifact class, then semantic reconstruction can begin in earnest; if the same filler holds across frames, phonetic testing becomes possible.

17.9 Outputs

- outputs/structural_reconstruction_summary.csv
- outputs/structural_reconstructions.csv
- outputs/structural_reconstruction_templates.csv
- outputs/structural_semantic_frames.csv
- outputs/phonetic_bootstrap_candidates.csv
- outputs/sign_proto_glosses.csv
- outputs/structural_frame_usage.csv
- outputs/structural_reconstruction_model.tex

18 Phonetic Bootstrap Testbench

18.1 Purpose

The project can now begin phonetic bootstrapping in a controlled sense. This does not mean that a sign has been read as a sound. It means that a reproducible testbench now assigns abstract variables to signs and extracts the inscriptions where future sound values, morphemes, titles, classifiers, or proper-name readings can be falsified.

The notation is deliberately conservative. A variable such as C01, T01, or R02 is not a phoneme. It is a placeholder for a sign whose function is constrained enough that a phonetic hypothesis can later be tested against every occurrence.

18.2 Method

The script `scripts/phonetic-bootstrap-testbench.py` combines:

- the 607 medium/high phonetic-readiness reconstructions from the structural model;
- functional proto-glosses for signs such as 820, 861, 817, 740, and 520;
- cross-frame filler reuse from the slot-paradigm model;
- minimal alternation pairs within the same structural frame.

Each sign receives a stable abstract variable. The prefix marks only the current functional class:

- C: classifier/title-like sign;
- T: terminal title/suffix-like sign;
- M: formula marker;
- A: slot modifier;
- R: possible root or unresolved sign.

18.3 Corpus-Wide Result

Metric	Value
Medium/high ready reconstructions	607
Abstract sign variables	338
Distinct reading units	627
Strong onomastic anchors	7
Cross-frame phonetic anchors	31
Affix/formula anchors	75
Prime-entity fillers	85
Abstract reconstructions exported	607
Minimal alternation tests	300

Table 40: Abstract phonetic bootstrap summary.

18.4 Variable Map

The strongest currently constrained variables are shown in Table 41. These are the variables that future proposed readings must explain first.

18.5 Strong Onomastic Anchors

The most important new class is the strong onomastic anchor: a prime-entity filler that appears in a semantically narrow slot, occurs across more than one site, and participates in minimal alternation evidence. These are the closest analogues, in our current data, to the proper-name strategy used in successful decipherments.

Sign	Variable	Current functional class
820	C01	classifier/title-like
861	C02	classifier/title-like
817	C03	classifier/title-like
740	T01	terminal title/suffix-like
520	T02	terminal title/suffix-like
235	M01	formula marker / slot modifier
390	M02	formula marker / slot modifier
002	A01	slot modifier
240	A02	slot modifier
176	R02	possible root or unresolved sign

Table 41: Selected abstract variables. These are not sound values.

Unit	Abstract	Role	Frame evidence	Score
176	R02	prime entity filler	002 740; <START> 741	0.790
032-220	A07-A09	prime entity filler	002 740; <START> 233	0.757
590-390	A08-M02	prime entity filler	002 740	0.737
240-482	A02-R16	prime entity filler	002 740	0.664
235-222	M01-R31	prime entity filler	002 740	0.623
240-176	A02-R02	prime entity filler	002 740	0.608
235-840-032	M01-R10-A07	prime entity filler	002 740	0.600

Table 42: First strong onomastic-style anchors for controlled phonetic testing.

18.6 Example Abstract Reconstructions

The output file `outputs/abstract_phonetic_reconstructions.csv` renders 607 inscriptions in abstract variables. A few examples:

- **L-41, Lothal:**
817-002-233-740-090 to C03-A01-A04-T01-A10.
The sign 233 is the prime entity filler inside 002 X 740.
- **M-708, Mohenjo-daro:**
140-390-002-061-740 to M07-M02-A01-R29-T01.
This also preserves the 002 X 740 entity slot.
- **M-1114, Mohenjo-daro:**
920-060-741-002-140-740 to M04-A06-A03-A01-M07-T01.
This is a longer prime-entity seal formula.

This is the first point where the project can speak of sentence-like reconstruction with a phonetic layer: the formula is structurally and semantically constrained, while the sound layer remains abstract.

18.7 Minimal Alternation Protocol

The file `outputs/phonetic_minimal_tests.csv` contains the top 300 same-frame alternations. These are essential because a proposed reading is not allowed to explain only one attractive inscription. It must survive the alternation set.

For example, if two fillers occupy the same 002|740 prime-entity slot and differ by one sign, then a proposed phonetic value for that sign must preserve the same frame. The alternation may indicate a phonemic contrast, an affix, a classifier, an allograph, a numeral, or a graphic modifier. The testbench does not decide which one is correct; it gives us the falsification grid.

18.8 Interpretation

The current working reconstruction pipeline is now:

STRUCTURAL FRAME -> SEMANTIC OPTIONS -> ABSTRACT PHONETIC VARIABLES -> MINIMAL TESTS

The next empirical breakthrough will probably not come from guessing a sound for a sign in isolation. It should come from taking a strong anchor such as 176, 590–390, or 235–222, collecting every artifact and image context where it appears, and asking whether one proposed semantic or phonetic value predicts the whole set.

18.9 Outputs

- outputs/phonetic_bootstrap_summary.csv
- outputs/phonetic_variable_map.csv
- outputs/phonetic_reading_units.csv
- outputs/abstract_phonetic_reconstructions.csv
- outputs/phonetic_minimal_tests.csv
- outputs/phonetic_bootstrap_testbench.tex

19 Semantic Context Triangulation

19.1 Purpose

The project can now begin semantic reconstruction in a constrained way. This section connects the structural and abstract phonetic models to corpus metadata: site, region, object type, material, and iconographic symbol labels. The goal is to discover whether a candidate reading unit behaves like a name, title, place, office, commodity, emblem, or administrative formula.

This is not translation. A context association such as “unit 176 is often found on bull-associated seals” does not mean that 176 reads “bull”. It means that any future reading of 176 must explain that context.

19.2 Method

The script `scripts/semantic-context-triangulation.py` uses the 607 medium/high phonetic-readiness reconstructions. For each slot filler and reading unit, it counts contextual distributions across:

- region and site;
- artifact type;
- material;
- iconographic symbol label;
- semantic frame.

For each unit/context pair, the model computes observed count, expected count, lift, log-lift, and an association score. The score rewards repeated support, purity within the unit’s occurrences, and lift above corpus baseline. The output prioritizes units already marked as strong onomastic anchors, cross-frame phonetic anchors, or affix/formula anchors.

19.3 Corpus-Wide Result

Metric	Value
Ready reconstructions tested	607
Reading units profiled	627
Strong onomastic units profiled	7
Usable iconography labels	482
Unknown iconography labels	125
Strong context associations	60
Moderate context associations	186
Symbol-linked semantic candidates	33
Semantic candidates exported	113

Table 43: Semantic context triangulation summary.

19.4 Prime-Entity Semantic Leads

The first strong semantic target remains the 002 X 740 prime-entity frame. The current best leads are:

- 240-482 (A02-R16): Harappa/tablet-heavy; likely administrative or institutional label.
- 176 (R02): Mohenjo-daro plus Lothal; bull/gaur seal context; possible name, office, lineage, or emblem-linked identity.
- 235-222 (M01-R31): bull seals with Chanh-daro concentration; emblem-linked name/title candidate.
- 032-220 (A07-A09): bull seals across sites; cross-site emblem/name candidate.
- 590-390 (A08-M02): steatite seals; bull-linked where known; name/title/place candidate.
- 240-176 (A02-R02): steatite seals; sparse but phonetic-ready candidate.
- 235-840-032 (M01-R10-A07): elephant/bull seals; sparse prime-entity candidate.

19.5 Important Non-Onomastic Leads

Some high-scoring units are not likely to be proper names. They may be titles, suffixes, object labels, or administrative formulae:

- 240-520 strongly clusters with Harappa tablets and bull-labeled contexts. It may be a terminal administrative expression rather than a personal name.
- 240-740-090 is associated with tablet contexts and a **Cros** symbol label. This makes it a strong object-class or administrative-ending candidate.
- 032 is notably Chanh-daro-heavy, suggesting a possible regional, scribal, or administrative specialization.
- 140 has a strong goat-labeled subgroup while also appearing in bull contexts, so it should be treated as a classifier/title/formula marker until image review resolves the context.

19.6 Interpretation

The project has now reached semantic proto-reconstruction. The safest current semantic template for many seal inscriptions is:

CLASSIFIER/TITLE + ENTITY SLOT + TERMINAL/FINAL

For the prime frame this is:

002 + ENTITY + 740

The entity may be a personal name, title, lineage, place, institution, commodity, or emblem-linked identity. The context model now lets us choose which possibility to test first for each filler.

19.7 Next Empirical Test

The most important next step is image-grounded validation. The corpus symbol labels are useful but not enough. For each strong anchor, we need to inspect the actual seal or inscription images and ask:

- Does the same unit consistently occur with the same iconographic motif?
- Does it remain stable across site, material, and object type?
- Do minimal alternations preserve the same iconographic and structural context?
- Does a proposed semantic reading predict all occurrences, including the awkward ones?

The immediate priority set is 176, 235-222, 032-220, 590-390, and 240-482. These give us the first real bridge from structure toward meaning.

19.8 Outputs

- `outputs/semantic_context_summary.csv`
- `outputs/anchor_context_profiles.csv`
- `outputs/anchor_context_correlations.csv`
- `outputs/semantic_reconstruction_candidates.csv`
- `outputs/iconography_semantic_matrix.csv`
- `outputs/semantic_context_triangulation.tex`

20 Language Hypothesis Testbench

20.1 Purpose

This section tests language and script hypotheses against the current structural, abstract phonetic, and semantic models. It does not claim decipherment. Its job is to separate three levels of claim:

- a script-function claim, such as restricted administrative or onomastic writing;
- a language-family compatibility claim, such as Dravidian-compatible suffixing structure;
- a true language identification claim, which would require validated phonetic readings.

The project is now strong enough to score the first two levels. It is not yet strong enough to make the third.

20.2 Method

The script `scripts/language-hypothesis-testbench.py` converts each hypothesis into explicit diagnostics and scores those diagnostics against existing outputs:

- structural formulae and semantic frames;
- abstract phonetic variables and minimal-pair evidence;
- semantic context associations with site, region, artifact type, material, and iconography;
- strong onomastic anchors in the 002 X 740 prime-entity frame.

The model penalizes all language-family hypotheses because no sign has yet received a validated sound value. This is intentional. Compatibility is not identification.

20.3 Current Result

Rank	Hypothesis	Score
1	Restricted administrative/onomastic script	0.731
2	Multilingual administrative ecology	0.689
3	Primarily non-linguistic emblem system	0.621
4	Dravidian-compatible linguistic layer	0.605
5	Logo-syllabic or logo-phonetic writing	0.585
6	Indo-Iranian/Indo-Aryan linguistic layer	0.541
7	Munda/Para-Munda linguistic layer	0.527

Table 44: Hypothesis ranking. Scores are compatibility scores, not proof.

The best overall model is currently a restricted administrative/onomastic script. The best language-family-compatible model is Dravidian-compatible, mostly because the corpus now shows strong suffix/final behavior and productive name/title-like slots. However, this is not enough to identify the language. Indo-Iranian/Indo-Aryan and Munda/Para-Munda remain plausible but less strongly supported by the current structural evidence.

20.4 Diagnostic Features

Feature	Score
Formula restriction	0.898
Context association	0.890
Usable iconography	0.794
Suffix/final structure	0.734
Regional differentiation	0.733
Prime entity structure	0.654
Phonetic readiness	0.583
Classifier/title structure	0.517

Table 45: Diagnostic evidence features.

The suffix/final score is the strongest reason Dravidian remains interesting. The classifier/title and context-association scores are the strongest reasons a language-neutral or multilingual administrative model remains stronger than any single-language claim.

20.5 What We Can Claim Now

- The inscriptions are highly compatible with restricted formulaic writing.
- The 002 X 740 frame is the best current partial-decipherment target.
- X is likely an entity slot: name, title, office, lineage, place, institution, commodity, or authority formula.
- Dravidian-compatible suffixing is currently the strongest language-family-compatible pattern.
- No language-family identification is justified until at least one phonetic or lexical value predicts all occurrences of a strong anchor.

20.6 Lexical Reading Gate

The file `outputs/lexical_reading_gate.csv` defines what kinds of lexical searches are allowed for each unit. For the strongest prime-entity anchors, lexical searches are limited to proper names, offices, titles, lineages, places, institutions, and emblem-linked identities. The model explicitly blocks shortcuts such as equating a sign with an animal name just because the seal has that animal.

The immediate reading gate for the top anchors is:

- 176: search only after all Mohenjo-daro and Lothal occurrences are image-checked.
- 235-222: test bull-linked name/title readings, but only if the Chanhudaro cluster holds visually.
- 032-220: test cross-site emblem/name readings across bull-seal contexts.
- 240-482: test Harappa administrative or institutional readings before personal-name readings.
- 590-390: test steatite seal name/title/place readings after image validation.

20.7 Interpretation

The project has not cracked the language yet, but it has narrowed the problem. The current best model is:

RESTRICTED ADMINISTRATIVE/ONOMASTIC SCRIPT

with a probable formula:

CLASSIFIER/TITLE + ENTITY + TERMINAL/SUFFIX

and a prime target:

002 + ENTITY + 740

The strongest next decipherment route is therefore not a global alphabet or syllabary. It is controlled proper-name and title testing, using image-grounded semantic anchors and minimal alternations.

20.8 Outputs

- `outputs/language_evidence_features.csv`
- `outputs/language_hypothesis_scores.csv`
- `outputs/lexical_reading_gate.csv`
- `outputs/decipherment_claims.csv`
- `outputs/language_hypothesis_summary.csv`
- `outputs/language_hypothesis_testbench.tex`

21 Anchor Dossier Model

21.1 Purpose

The project now needs case files rather than broader rankings. This section turns the strongest candidate units into anchor dossiers: each dossier gathers occurrence rows, context, minimal alternations, component reuse, and allowed lexical search classes. The goal is controlled partial decipherment.

This remains pre-phonetic. The output proposes reading classes such as “emblem-linked name/title candidate” or “administrative label candidate”, not sound values.

21.2 Method

The script `scripts/anchor-dossier-model.py` combines:

- semantic reconstruction candidates;
- lexical reading gates;
- structural reconstructions and slot fillers;
- phonetic minimal-pair tests;
- abstract sign variables.

The model keeps all strong onomastic anchors and strong-context cross-frame anchors. It then asks whether each anchor is a stable unit, a compound, or a sequence of reusable morpheme-like components.

21.3 Corpus-Wide Result

21.4 Highest-Priority Dossiers

- 240-482: institutional/admin label candidate; Harappa-heavy; mostly tablet context.
- 176: emblem-linked name/title candidate; appears as 176 and in 240-176.
- 235-222: emblem-linked name/title candidate; bull context and Chanhudaro concentration.
- 032-220: emblem-linked name/title candidate; bull context across Mohenjo-daro and Harappa.
- 590-390: emblem-linked name/title candidate; steatite seal context across three sites.

Metric	Value
Anchor dossiers	11
Strong onomastic dossiers	7
Cross-frame dossiers	4
Anchor occurrence rows	59
Direct minimal tests	49
Component minimal tests	277
Anchor components	15
Root-like components	2
Prefix-like components	3
Suffix-like components	4

Table 46: Anchor dossier summary.

- 240-176: controlled lexical candidate; may contain the same root-like 176 with 240 as a qualifier.
- 235-840-032: controlled lexical candidate; shares 235 and 032 with other anchors.

21.5 Morpheme-Like Component Probe

The most important new result is component reuse inside the strongest anchors:

- 176 appears as a standalone prime-entity filler and as the final component in 240-176. This is the cleanest root-like entity component so far.
- 032 appears as a standalone cross-frame unit and inside 032-220 and 235-840-032. This makes it root-like or subtype-marking, not merely decorative.
- 240 appears initially in 240-482, 240-176, and 240-740-090. It is currently the strongest prefix/title/classifier-like component in the anchor set.
- 235 appears initially in 235-222 and 235-840-032; it remains formula-marker or title-prefix-like.
- 220, 222, 390, 482, and 840 are lexical-root or final-qualifier candidates that need image-grounded validation.

21.6 Working Partial-Reading Classes

The strongest controlled reading classes are now:

$$240 + \text{ENTITY} \quad \text{and} \quad 235 + \text{ENTITY}$$

This does not mean 240 or 235 has a phonetic value. It means they behave like reusable entity qualifiers in the prime-entity zone. If future image review confirms that 176 and 240-176 share a semantic context, then 240 becomes a serious prefix/title/classifier candidate and 176 becomes a serious root-like candidate.

21.7 Interpretation

This is the first stage where the project can make a morpheme-like statement:

$$\text{QUALIFIER} / \text{TITLE} / \text{CLASSIFIER} + \text{ENTITY STEM}$$

For example, 240-176 is no longer just a two-sign filler. It is a testable compound: 240 may qualify or classify the root-like 176. The same logic applies to 235-222 and 235-840-032.

This is a meaningful step toward language, because language identification will require reusable segments that behave predictably across compounds. We now have a small set of such segments, but they remain abstract until image context and minimal-pair constraints validate them.

21.8 Outputs

- `outputs/anchor_dossier_summary.csv`
- `outputs/anchor_dossiers.csv`
- `outputs/anchor_occurrence_evidence.csv`
- `outputs/anchor_minimal_evidence.csv`
- `outputs/anchor_component_roles.csv`
- `outputs/anchor_component_edges.csv`
- `outputs/anchor_reading_hypotheses.csv`
- `outputs/anchor_dossier_model.tex`

22 Constraint-Solver Reading Layer

22.1 Purpose

The project has reached the point where speed depends on automatic rejection. This section adds a constraint-solver layer that converts anchor dossiers into ranked reading skeletons. The solver is deliberately pre-phonetic: it can accept a structural reading such as **QUALIFIER/TITLE** + **ENTITY STEM**, but it cannot assign a sound value, language-family value, or lexical translation.

This is the safest fast path toward decipherment. It lets us scale from a few promising anchors to many hypotheses while keeping each proposal falsifiable.

22.2 Method

The script `scripts/constraint-solver-reading-layer.py` combines:

- anchor dossiers and component roles;
- occurrence-level context rows;
- minimal-pair and component-alternation evidence;
- lexical reading gates;
- structural frames and semantic frame labels.

Each candidate is tested against seven constraints: component order, context support, minimal-pair support, frame stability, lexical-gate compliance, phonetic restraint, and functional-role availability. The output is a ranked list of skeleton readings, not deciphered sentences.

Metric	Value
Candidate anchor skeletons	11
Accepted/promising skeletons	6
Exploratory skeletons	5
Hard-constraint holds	0
Clause reconstructions exported	59
Phonetic readings assigned	0

Table 47: Constraint-solver summary.

22.3 Solver Result

The strongest accepted skeleton is:

240-482 = QUALIFIER/TITLE PREFIX + ENTITY STEM

The most important supporting chain is:

176 = ENTITY STEM 240-176 = QUALIFIER/TITLE PREFIX + ENTITY STEM

This is not a translation. It is a controlled structural claim: 240 behaves like a reusable qualifier/title/classifier-like component, while 176 behaves like a reusable root-like entity component.

22.4 Highest-Ranked Reading Skeletons

Unit	Skeleton	Tier	Score
240-482	QUALIFIER_TITLE_PREFIX + ENTITY_STEM	A	0.966
240-740-090	ADMINISTRATIVE_OBJECT_FORMULA	A	0.951
032	TERMINAL_TITLE_OR_OBJECT_ENDING	A	0.882
176	ENTITY_STEM	B	0.849
240-176	QUALIFIER_TITLE_PREFIX + ENTITY_STEM	B	0.783
590-390	QUALIFIER_TITLE_PREFIX + FINAL_QUALIFIER	B	0.757

Table 48: Highest-ranked pre-phonetic reading skeletons.

22.5 Approximate Distance to Full Decipherment

Full decipherment remains far. The project is no longer at zero: we have corpus control, structural frames, abstract variables, semantic context scores, anchor dossiers, and now constrained reading skeletons. But a strict decipherment requires validated sound values and a language identification, and those remain mostly unearned.

As an approximate evidence-maturity estimate:

The honest answer is therefore: we may be roughly halfway toward a serious structural decipherment testbench, but only around a tenth of the way toward strict full decipherment. The fastest safe path is not to guess a language now, but to build a falsifiable chain:

ANCHORS -> COMPONENTS -> PROPER NAMES/TITLES -> PHONETIC TESTS

Subproblem	Maturity
Corpus and provenance control	0.70
Structural grammar	0.48
Morpheme segmentation	0.27
Semantic anchoring	0.30
Phonetic values	0.03
Language identification	0.06
Strict full decipherment	0.10

Table 49: Heuristic maturity estimates. These are not probabilities of final success.

22.6 Next Research Decision

The next milestone should expand the solver from the eleven anchor units to their nearest minimal-pair neighbors. This directly attacks the phonetic problem: if 240-176, 240-482, and related alternants preserve frame identity while substituting only one component, then the project can begin testing whether the contrast behaves like a title, stem, suffix, numeral, allograph, or phonetic alternation.

22.7 Outputs

- `outputs/constraint_solver_summary.csv`
- `outputs/constrained_reading_candidates.csv`
- `outputs/constraint_violations.csv`
- `outputs/morpheme_slot_assignments.csv`
- `outputs/reconstructed_clause_frames.csv`
- `outputs/decipherment_progress_estimate.csv`
- `outputs/constraint_solver_model.tex`

23 Minimal-Pair Neighbor Expansion

23.1 Purpose

The constraint solver gave us a small set of accepted reading skeletons. This section expands those anchors into their nearest same-frame neighbors. The aim is to identify which contrasts are likely to be title/classifier differences, entity-stem differences, terminal/suffix differences, numeral/allograph differences, or merely role-boundary controls.

This is the fastest safe route toward phonetics because it asks where a sound value could eventually be forced by contrast. It also prevents false progress: if two signs do not yet occupy the same functional role, the model blocks them from being used as phonetic evidence.

23.2 Method

The script `scripts/minimal-pair-neighbor-expansion.py` combines:

- constrained anchor skeletons;
- morpheme slot assignments;

- full same-frame minimal-pair tests;
- structural occurrence profiles for neighboring fillers;
- abstract sign roles from the phonetic variable map.

It separates exact anchor-neighbor tests from component-neighbor tests. Exact tests involve the whole anchor unit, such as 240-482 versus 240-176. Component tests involve a reusable component inside the anchor, such as 240 alternating with another initial component.

23.3 Corpus-Wide Result

Metric	Value
Anchor neighborhoods expanded	11
Neighbor tests	333
Exact anchor-neighbor tests	49
Component neighbor tests	284
Component contrast groups	156
Probe queue items	30

Table 50: Minimal-pair neighbor expansion summary.

The largest class is not phonetic evidence but role-boundary control. This is important: the model found many comparisons that look tempting but are not safe for phonetic inference because the signs may be serving different roles.

23.4 Best Current Morphology Probes

The strongest morphology probes are:

- 240 deletion/addition in 002|<END> contexts, especially through the neighborhoods of 240-482, 240-176, and 240-740-090. This keeps 240 as the leading qualifier/title/classifier candidate.
- 240 versus 235 in initial-slot contexts such as <START>|233. This tests whether the two are competing title/classifier/formula openers.
- 740 versus 520 in terminal frames such as 233|<END>. This tests a terminal/suffix or numeral/allograph contrast.

23.5 Best Current Stem and Phonetic Probes

The cleanest exact stem probes are:

- 176 versus 048 in 002|740, combined count 20.
- 176 versus 061 in 002|740, combined count 17.
- 240-482 versus 240-904 in 002|740, combined count 11.
- 240-176 versus 240-482 in 002|740, combined count 10.
- 240-482 versus 240-773 in 002|740, combined count 10.

These are still not readings. They are phonetic-probe candidates. Before any sound value is proposed, each pair must survive image review and context comparison: the changed component must explain the difference without breaking the frame.

23.6 Important Negative Result

032 produces many high-count same-frame contrasts, but the solver now treats most of them as role-boundary controls. This is a useful restraint. 032 appears in terminal, initial, and compound contexts, so it cannot yet be used as direct phonetic evidence. Its first job is to help us understand slot function, not sound.

23.7 Interpretation

We now have a concrete bridge from structural decipherment toward phonetic decipherment:

ANCHOR SKELETON → MINIMAL NEIGHBORHOOD → CONTROLLED CONTRAST → PHONETIC
PROBE

The best immediate route is to image-check the 176 stem neighborhood and the 240 + STEM neighborhood. If those contexts separate cleanly, the project can begin testing whether 176, 048, 061, 482, 904, and 773 behave like names, titles, places, institutions, or other entity stems.

23.8 Outputs

- `outputs/neighbor_expansion_summary.csv`
- `outputs/expanded_anchor_neighborhoods.csv`
- `outputs/neighbor_reading_tests.csv`
- `outputs/component_contrast_tests.csv`
- `outputs/phonetic_probe_queue.csv`
- `outputs/minimal_pair_neighbor_expansion.tex`

24 Contrastive Probe Validation

24.1 Purpose

This section moves from a probe queue to controlled decipherment constraints. The previous model listed promising minimal-pair neighborhoods. This validation layer asks which probes can already be promoted to grammar constraints, which can become phonetic probes after image checking, and which must remain controls.

The key rule remains strict: no sound value, lexical translation, or language-family claim is allowed unless the proposed value predicts all occurrences of the relevant contrast.

24.2 Method

The script `scripts/contrastive-probe-validation.py` combines:

- the top thirty minimal-pair probes;
- exact occurrence profiles from structural reconstructions;
- same-frame minimal-pair counts;
- component roles from the morpheme assignment table;
- site, object, material, symbol, and semantic-frame distributions.

The validator separates three levels of result:

- grammar constraints, which can be used now;
- stem-semantic probes, which are promising but occurrence-limited;
- phonetic-ready probes, which still require image checking before any abstract sound contrast is proposed.

24.3 Validation Result

Metric	Value
Probe validations	30
Morphology constraints	11
Terminal constraints	1
Phonetic-ready after image check	1
Stem-semantic probes	3
Full-neighborhood role-boundary controls	108
Stem lattice edges	7

Table 51: Contrastive probe validation summary.

24.4 First Phonetic-Ready Edge

The first probe to reach phonetic-ready status after image checking is:

240-482 :: 240-904

in the frame:

002|740

The controlled interpretation is:

240 + STEM

where 482 and 904 occupy the same second-position stem slot. The current evidence is still pre-phonetic: it licenses a future contrast test, not a sound value.

24.5 Stem Lattice

The strongest stem-lattice edges are:

Edge	Probe	Status	Score
176:048	240-176 :: 048	Stem-semantic probe	0.822
482:904	240-482 :: 240-904	Phonetic-ready after image check	0.819
176:048	176 :: 048	Stem-semantic probe	0.788
176:061	176 :: 061	Stem-semantic probe	0.750
176:482	240-176 :: 240-482	Hold for more occurrences	0.698
482:773	240-482 :: 240-773	Hold for more occurrences	0.696

Table 52: Top stem-lattice contrasts.

24.6 Grammar Constraints Now Licensed

The validation layer licenses the following grammar constraints:

- 240 behaves as a qualifier/title/classifier-like component in several neighborhoods.
- 240 and 235 compete in title/classifier/formula-opener environments.
- 740 and 520 form a terminal/suffix contrast family.
- 032 must remain blocked for phonetics because it creates many role-boundary controls.

24.7 Controlled Claim Ledger

The current claim ledger now contains six controlled claims:

- C-240-QUALIFIER: supported morphology constraint.
- C-176-STEM: stem-semantic probe.
- C-482-STEM-SLOT: emerging stem lattice.
- C-740-520-TERMINAL: terminal constraint.
- C-032-CONTROL: blocked for phonetics.
- C-LANGUAGE-GATE: language identification remains blocked.

24.8 Interpretation

This is a meaningful step closer to decipherment because the project now has its first phonetic-ready contrast candidate:

482 :: 904

inside:

240 + STEM

The next safe step is visual validation of every 240-482 and 240-904 occurrence. If the visual and contextual evidence holds, the project can begin testing an abstract contrast between two stem values. That would still not be a language reading, but it would be the first responsible opening toward phonetic decipherment.

24.9 Outputs

- `outputs/contrastive_probe_validation_summary.csv`
- `outputs/validated_probe_results.csv`
- `outputs/probe_occurrence_contexts.csv`
- `outputs/stem_contrast_lattice.csv`
- `outputs/controlled_decipherment_claims.csv`
- `outputs/contrastive_probe_validation.tex`

25 Breakthrough Target Portfolio

25.1 Purpose

This section identifies the targets where a successful resolution would make later phases disproportionately faster. The goal is to avoid flat exploration. A high-value target is not merely interesting; it should unlock many downstream tests.

25.2 Method

The script `scripts/breakthrough-target-prioritizer.py` combines:

- validated probe results;
- component contrast groups;
- the stem contrast lattice;
- sign roles and ready-text counts;
- role-boundary controls and blocked phonetic comparisons.

It scores targets by downstream unlock value: number of affected probes, corpus coverage, validated grammar constraints, role-boundary risk, and dependency links to later phases.

25.3 Portfolio Summary

Metric	Value
Breakthrough targets ranked	9
Dependency edges	9
Action-plan items	7

Table 53: Breakthrough target summary.

25.4 Highest-Leverage Targets

Rank	Target	Type	Score
1	240	Grammar keystone	1.000
2	740::520	Terminal keystone	0.828
3	176/482/904/048/061/773	Stem lattice keystone	0.788
4	002 X 740	Frame keystone	0.737
5	482::904 inside 240+STEM	Phonetic keystone	0.726
6	176 with 048/061	Stem-family keystone	0.718
7	240::235	Formula boundary keystone	0.695
8	032 role split	Blocker removal	0.661

Table 54: Highest-leverage decipherment targets.

25.5 Interpretation

The top target is 240. If 240 can be stabilized as a qualifier, title, or classifier-like component, then 240-482, 240-176, and 240-740-090 become part of one controlled grammar problem instead of separate curiosities. That makes the stem contrasts easier to test.

The second target is 740::520. These signs affect the terminal side of many inscriptions. Solving this contrast would constrain whether the right edge is suffixal, titular, numerical, allographic, or formulaic.

The first phonetic keystone remains 482::904 inside 240+STEM. It is not ranked first because it depends on the grammar environment: if 240 is not understood well enough, the phonetic test floats without an anchor.

25.6 Action Plan

The recommended breakthrough order is:

1. Promote all validated 240 morphology constraints into a grammar model.
2. Model 740::520 as competing terminal functions.
3. Build a stem-lattice consistency test for 176, 482, 904, 048, 061, and 773.
4. Treat 002|740 as a formal test environment and score every X filler.
5. Image-check all 240-482 and 240-904 occurrences.
6. Verify every 048 and 061 occurrence in 002|740.
7. Model 240::235 as a rival opener contrast.

25.7 Dependency Logic

The portfolio has a dependency chain:

240 GRAMMAR -> 240+STEM -> 482::904 -> STEM LATTICE -> PHONETIC TESTS

and a second chain:

740::520 -> 002 X 740 FRAME -> ENTITY-SLOT GRAMMAR

The main blocker is 032. It creates many role-boundary controls. Until its functions are split, it should protect the project from false phonetic comparisons rather than drive new readings.

25.8 Outputs

- outputs/breakthrough_target_summary.csv
- outputs/breakthrough_targets.csv
- outputs/breakthrough_dependency_edges.csv
- outputs/breakthrough_action_plan.csv
- outputs/breakthrough_target_portfolio.tex

26 Grammar Keystone 240 Model

26.1 Purpose

The breakthrough portfolio ranked 240 as the highest-leverage target. This section executes that target. It tests whether 240 should be treated as a reusable grammar operator, an ordinary stem, a formula boundary sign, a terminal marker, or a mixed sign. The model is still pre-phonetic: it licenses controlled environments, not sound values.

26.2 Summary

Metric	Value	Note
240 token occurrences	354	texts=354
Position profile	Medial:234; Initial:96; Final:16; Standalone:8	Reading-order positions in cleaned corpus.
Distinct signs after 240	69	520:46; 740:25; 233:24; 636:19; 482:15
Distinct signs before 240	28	002:83; 235:45; 798:20; 060:16; 220:15
Observed local frames	182	002 520:29; 235 740:15; <START> 636:10; <START> 705:9; <START> 233:9
240+X slot groups	54	18 frame-locked groups at score ≥ 0.70
Controlled 240+STEM lanes	3	Can feed later phonetic tests after image checks.
Validated 240 morphology constraints	8	Promoted to grammar model only.
Leading 240 model	Qualifier/title/classifier prefix operator	score=0.986; status=LEADING MODEL

Table 55: Summary of the 240 grammar-keystone test.

26.3 Rival Models

Model	Score	Status
Qualifier/title/classifier prefix operator	0.986	LEADING MODEL
Polyfunctional mixed sign	0.615	CONTROL RISK
Formula opener/boundary marker	0.463	SECONDARY FUNCTION
Terminal/suffix/numeral value	0.060	LOW SUPPORT
Ordinary lexical stem	0.012	WEAKENED

Table 56: Rival grammatical models for 240.

26.4 240+X Families

26.5 Frame Identity Tests

26.6 Decision

The leading model treats 240 as a qualifier/title/classifier-like operator. This does not decipher 240; it constrains it. The working notation should therefore keep 240 as an abstract operator, OP240, while the following position is tested as the stem-like slot.

Post240	Count	Top frame	Lock	Lead
520	33	002 <END>	0.800	Prime 240+X environment; control grammar before treating X as a stem.
<NONE>	31	<START> 233	0.476	240 occurs as a complete filler; treat as possible standalone operator/control.
740	21	235 <END>	0.843	Frame-locked 240 compound; useful grammar control.
482	16	002 740	0.933	Prime 240+STEM environment; use as controlled stem lane.
415	12	002 740	0.817	Mixed frame evidence; keep as lower-confidence 240 context.
636	11	031 <END>	0.579	Mixed frame evidence; keep as lower-confidence 240 context.
220	7	002 <END>	0.739	Mixed frame evidence; keep as lower-confidence 240 context.
642	7	060 <END>	0.436	Mixed frame evidence; keep as lower-confidence 240 context.
904	7	002 740	0.751	Prime 240+STEM environment; use as controlled stem lane.
603	6	060 <END>	0.533	Mixed frame evidence; keep as lower-confidence 240 context.

Table 57: Highest-yield 240+X families.

Post240	With	Bare	Overlap	Result
176	4	14	0.571	240 stem lane
482	16	3	0.333	240 stem lane
520	33	42	0.132	240 grammar control
904	7	7	0.286	240 stem lane
220	7	23	0.261	mixed/low evidence
740	21	34	0.261	frame-locked; needs bare control
806	2	10	0.300	mixed/low evidence
235	3	35	0.143	mixed/low evidence
773	5	1	0.000	mixed/low evidence
233	4	13	0.481	mixed/low evidence

Table 58: Tests of whether 240+X preserves enough frame identity to become a controlled environment.

The practical consequence is large: 240-482, 240-904, 240-176, and related forms can now be tested as one grammar family. The first phonetic lane remains conditional on image review, but it no longer floats without structure.

26.7 Acceleration Queue

Priority	Action	Output
1	Freeze 240 as a grammar operator in the solver	Update constrained readings so 240 is an operator label, not a sound-bearing value.
2	Promote the first phonetic-grade 240+STEM lane	Image-check and context-check 240-482 against 240-904 as one controlled contrast.
3	Use 176 as the bridge/control stem	Compare bare 176, 240-176, 048, and 061 inside the prime entity frame.
4	Use frame-locked 240 compounds as controls	Generate a 240+STEM contrast matrix for 482, 904, 176, 773, and high-count followers.
5	Split 240 from 235 before lexical claims	Build a rival-opener model: 240+X versus 235+X and X+terminal contexts.
6	Block sound values for 240	Keep 240 as OP240 in abstract reading skeletons.

Table 59: Immediate actions unlocked by the 240 grammar model.

26.8 Outputs

- `outputs/grammar_keystone_240_summary.csv`
- `outputs/grammar_keystone_240_occurrences.csv`
- `outputs/grammar_keystone_240_compound_families.csv`
- `outputs/grammar_keystone_240_preservation_tests.csv`
- `outputs/grammar_keystone_240_operator_models.csv`
- `outputs/grammar_keystone_240_action_queue.csv`

27 Stem Lattice Keystone Model

27.1 Purpose

The 240 grammar model gives us a controlled environment. This section turns that environment into a stem lattice: a ranked space of stem signs and stem contrasts in the prime entity frame. The result is not a decipherment, but it is a move toward one: it identifies which contrasts can now be tested repeatedly without changing the surrounding grammar.

27.2 Summary

27.3 Stem Profiles

27.4 Contrast Matrix

27.5 Structural Templates

27.6 Decision

The current fastest lane is 240-482 :: 240-904. It is the first contrast that is both structurally controlled by OP240 and already marked as phonetic-ready after image checking. The bridge-control lane is 176 :: 048 :: 061, because 176 has both bare and OP240-prefixed evidence while 048 and 061 are strong bare controls in the same prime entity frame.

Metric	Value	Note
Stem profiles	103	low-evidence stem:71; bare prime control:15; OP240/bare bridge:14; phonetic-ready OP240 stem:2; OP240/bare bridge stem:1
Pairwise contrasts	120	occurrence-limited contrast:89; phonetic-lane neighbor:26; bridge stem control:3; primary phonetic lane:1; stem-semantic control:1
Top stem	482	phonetic-ready OP240 stem; score=1.000
Top pair	482::904	primary phonetic lane; score=1.000
Primary phonetic lane	482::904	Only after image/context validation; no sound value assigned.
Bridge controls	176; 048; 061	Tests whether OP240 preserves identity across bare and prefixed forms.

Table 60: Stem lattice keystone summary.

Stem	Role	Score	OP240	Bare
482	phonetic-ready OP240 stem	1.000	16	2
904	phonetic-ready OP240 stem	1.000	6	5
176	OP240/bare bridge stem	0.898	4	34
415	OP240/bare bridge	0.620	8	23
017	OP240/bare bridge	0.508	2	16
231	OP240/bare bridge	0.489	1	21
840	OP240/bare bridge	0.458	2	10
048	bare prime control	0.443	0	23
061	bare prime control	0.432	0	30
350	OP240/bare bridge	0.426	1	10

Table 61: Highest-ranked stem signs in the prime entity environment.

Pair	Class	Score	Validation
482::904	primary phonetic lane	1.000	0.819
482::176	bridge stem control	0.898	0.698
176::048	stem-semantic control	0.807	0.822
176::061	bridge stem control	0.708	0.750
904::350	phonetic-lane neighbor	0.667	0.000
904::415	phonetic-lane neighbor	0.666	0.000
482::350	phonetic-lane neighbor	0.663	0.000
482::415	phonetic-lane neighbor	0.663	0.000
482::772	phonetic-lane neighbor	0.663	0.000
904::017	phonetic-lane neighbor	0.661	0.000

Table 62: Highest-ranked stem contrasts.

Template	Reading	Use
002 OP240 STEM 740	prime entity frame with OP240-qualified stem	Main lane for 482, 904, and 176 before phonetic assignment.
002 STEM 740	bare prime entity stem	Tests whether OP240 preserves stem identity or changes function.
OP240 + 482 :: OP240 + 904	first phonetic-grade contrast lane after image review	Do not assign sound; use as the first controlled contrast to image-check.
176 :: 048 :: 061	bare/bridge semantic stem controls	Semantic and iconographic controls for the prime entity slot.

Table 63: Reusable structural templates produced by the stem lattice.

The reading discipline remains strict: OP240 is a grammatical operator, STEM is an abstract stem slot, and no Dravidian, Indo-Aryan, Munda, or other phonetic value is assigned yet.

27.7 Outputs

- `outputs/stem_lattice_keystone_summary.csv`
- `outputs/stem_lattice_keystone_profiles.csv`
- `outputs/stem_lattice_keystone_pair_matrix.csv`
- `outputs/stem_lattice_keystone_templates.csv`
- `outputs/stem_lattice_keystone_model.tex`

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